

```

1  %-----
2  % Nash Bargaining
3  %-----
4
5  clear
6  clc
7
8  %Printing the outputs as a text file if ==1
9  output_file=0;
10
11 %Problem Set-up
12 fun=@(x) -x(1)*x(2); %negative
13
14 A=[];
15 b=[];
16 Aeq=[];
17 beq=[];
18 lb=[0; 0];
19 ub=[];
20 nonlcon = @const; %non-linear constraint defined below
21 x0=[0.1; 0.1];
22
23 %fmincon
24 [x,fval,exitflag,output,lambda,grad,hessian] = fmincon(fun,x0,A,b,Aeq,beq,lb,ub,nonlcon);
25
26 if output_file==1
27     dfile="Nash_bargaining_fmincon.txt";
28     if exist(dfile, 'file')
29         delete(dfile);
30     end
31     diary(dfile)
32     diary on
33 end
34
35 %Output
36 fprintf('Nash Bargaining Solution: %.4f,%.4f\n',x(1),x(2));
37
38 if output_file==1
39     diary off
40 end
41
42 %Plot
43 slope_const=-x(1)/x(2); %The slope of the constraint - x1/x2
44 [x1, x2] = meshgrid(linspace(0, 1, 100), linspace(0, 1, 100));
45 figure();
46 hold on;
47 grid on;
48 contour(x1, x2, x1.^2 + x2.^2, [1, 1], 'LineWidth', 1, 'LineColor', 'k'); % constraint
49 contour(x1, x2, x1.*x2, [(-fval), (- fval)], 'LineWidth', 1, 'LineColor', 'b');
50 contour(x1, x2, x1.*x2, [0.75*(-fval), 1.25*(-fval)], 'LineWidth', 1, 'LineColor', 'k',
    'LineStyle', '--');
51 plot(x1(1,:), slope_const*(x1(1,:)-x(1))+x(2),'k','LineWidth', 1);
52 plot(x(1),x(2),'ro','LineWidth', 1);
53 xlabel('$x_{1}$','Interpreter','latex','FontSize',12);
54 ylabel('$x_{2}$','Interpreter','latex','FontSize',12);
55 title('Nash Bargaining Solution','Interpreter','latex','FontSize',14);
56 xlim([0,1]);
57 ylim([0,1]);

```

```
58
59 %Constraint
60 function [c,ceq] = const(x)
61 c = [x(1)^2+x(2)^2-1];
62 ceq = [];
63 end
64
```